OFFLINE NETWORK RESILIENCE PRINTABLE QUICK-CARDS

Cut, Laminate, and Distribute. No Internet Required.

Version 1.0 | 2026 | CC BY-SA 4.0 | 12 Cards Included

DESIGN NOTE:

Each card is designed to print on a half-sheet (5.5

″

×

4.25

″

) or standard index card. Print double-sided, laminate, and distribute. Card borders are indicated by a dashed-line box on each card. Cut along the dashed border after printing.

PRINTING INSTRUCTIONS

1. Print this document double-sided on standard 8.5

″

×

11

"

paper (two cards per sheet).

2. Align front and back carefully before cutting along the dashed borders.

3. Laminate each card at 3–5 mil thickness for field durability.

4. Distribute to all team members. Store one set in every go-box.

Tip:

Print at 100% scale — do not scale to fit. Use portrait orientation.

QUICK-CARD 1 — RADIO NET CHECK-IN PROCEDURE

QUICK-CARD 1 ▶ FRONT

RADIO NET CHECK-IN

When net control calls: "[Your call sign], go ahead"

YOUR RESPONSE:

1. State your **call sign**
2. State your **location** (grid square or neighborhood)
3. State your **status:** NORMAL / TRAFFIC / EMERGENCY
4. Wait for **acknowledgment**

If EMERGENCY:

Say

"BREAK BREAK BREAK"

+ your call sign + nature of emergency. Do not wait for net control to call you.

QUICK-CARD 1 ◀ BACK

PHONETIC ALPHABET

A = Alpha	B = Bravo	C = Charlie	D = Delta
E = Echo	F = Foxtrot	G = Golf	H = Hotel
I = India	J = Juliet	K = Kilo	L = Lima
M = Mike	N = November	O = Oscar	P = Papa
Q = Quebec	R = Romeo	S = Sierra	T = Tango
U = Uniform	V = Victor	W = Whiskey	X = X-ray
Y = Yankee	Z = Zulu		

QUICK-CARD 2 — EMERGENCY FREQUENCY REFERENCE

QUICK-CARD 2 ▶ FRONT

EMERGENCY FREQUENCIES — US

Band	Frequency	Use
NOAA Weather	162.400–162.550 MHz	Weather alerts
2m Calling	146.520 MHz	Amateur simplex
GMRS Ch 20	462.675 MHz	GMRS emergency
FRS Ch 1	462.5625 MHz	FRS general
CB Ch 9	27.065 MHz	CB emergency
Marine 16	156.800 MHz	Maritime distress
Aircraft Guard	121.500 MHz	Air emergency (RX only)
Meshtastic (US)	915 MHz LoRa	Mesh data

QUICK-CARD 2 ◀ BACK

LOCAL FREQUENCIES (FILL IN)

Field	Value
Repeater 1	_____ Tone: _____
Repeater 2	_____ Tone: _____
Simplex	_____
Net Time / Day	_____
Net Control	_____
Club / Group	_____

QUICK-CARD 3 — MESHTASTIC QUICK SETUP

QUICK-CARD 3 ▶ FRONT

MESHTASTIC SETUP IN 5 STEPS

5. Download Meshtastic app (Android / iOS) or use **flasher.meshtastic.org**
6. Flash firmware to device (TTGO T-Beam, Heltec, etc.)
7. Open app → pair via Bluetooth
8. Set: **Region = US**, Channel = **LongFast** (default)
9. Set device name → **You're on the mesh!**

CHANNEL SECURITY

- **Default:** No encryption (public)
- **Add PSK:** Settings → Channels → Add encryption key
- Share key **only** with trusted members

QUICK-CARD 3 ◀ BACK

MESHTASTIC TROUBLESHOOTING

Problem	Fix
No connection?	Restart device + app
No messages received?	Check region setting (must match)
Poor range?	Elevate antenna above obstacles
Battery draining fast?	Enable router role power settings
Can't pair?	Clear Bluetooth cache, re-pair

Default channel: LongFast **Frequency (US):** 906.875 MHz
Max message size: 228 bytes

QUICK-CARD 4 — SALUTE REPORT FORMAT

QUICK-CARD 4 ▶ FRONT

SALUTE REPORT — EMERGENCY INFORMATION

Use this format when reporting an incident over radio or in writing.

Element	Question to Answer
S — SIZE	How many people / vehicles involved?
A — ACTIVITY	What is happening?
L — LOCATION	Where exactly? (address, grid, landmark)
U — UNIT/UNIFORM	Who are they? (identifying features)
T — TIME	When did you observe this?
E — EQUIPMENT	What resources / tools are present?

QUICK-CARD 4 ◀ BACK

SAMPLE SALUTE REPORT

S:

Approximately 12 people

A:

Evacuating a flooded building

L:

Corner of Main St and Oak Ave

U:

Mix of civilians, 2 in firefighter gear

T:

Observed at 14:30 local

E:

One boat, several life vests, flashlights

Transmit clearly. **Pause between each element.** Wait for acknowledgment before proceeding.

QUICK-CARD 5 — VISUAL SIGNAL REFERENCE

QUICK-CARD 5 ▶ FRONT

COMMUNITY VISUAL SIGNALS

COLOR FLAGS / TAPE — WINDOWS OR FENCES:

Color	Meaning
GREEN	All clear — we are OK
YELLOW	Caution — need information or check-in
RED	HELP NEEDED — emergency assistance required
ORANGE	Medical need — injury or illness
BLUE	Have resources to share (water, food, supplies)
BLACK	Do not enter — hazard present
WHITE	Evacuated — structure unoccupied

QUICK-CARD 5 ◀ BACK

CHALK / DOOR & LIGHT SIGNALS

Search & Rescue Door Markings

X in box	Structure has been searched
Top quadrant	Date and time of search
Left quadrant	Hazards present
Right quadrant	Number of survivors
Bottom quadrant	Number of deceased

Mirror / Light Signals

3 flashes	Distress signal
Steady light	All clear / presence signal
Long-short alternating	Need assistance

QUICK-CARD 6 — INFORMATION RUNNER PROTOCOL

QUICK-CARD 6 ▶ FRONT

INFORMATION RUNNER — FIELD PROTOCOL

BEFORE DEPARTING

☐ Receive message **in writing** (never rely on memory alone)

☐ Confirm destination and **recipient name**

☐ Note priority level: ROUTINE / URGENT / EMERGENCY

☐ Log departure time and route

IN TRANSIT

☐ Travel in pairs when possible

☐ Use pre-planned routes only

☐ Do not detour without authorization

☐ If intercepted: **do not surrender message** without confirmation code

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QUICK-CARD 6 ◀ BACK

ON ARRIVAL & MESSAGE FORMAT

ON ARRIVAL

☐ Identify yourself and origin

☐ Request confirmation code from recipient

☐ Hand over message; receive **signed receipt**

☐ Log arrival time; wait for reply if instructed

☐ Log return departure

MESSAGE FORMAT

FROM:	**TIME:**
TO:	**PRIORITY:**
MESSAGE:	
SIGNATURE:	

QUICK-CARD 7 — MESH NODE HEALTH CHECK

QUICK-CARD 7 ▶ FRONT

MESH NODE — DAILY / WEEKLY CHECK

VISUAL CHECKS

☐ Antenna is secure and undamaged

☐ Enclosure is sealed; no water intrusion

☐ Solar panel is unobstructed and clean

☐ No physical damage to cables

ELECTRONIC CHECKS

☐ Device is powered on (indicator LED active)

☐ Battery voltage: min **12.0V** (lead-acid) / **3.0V/cell** (LiFePO4)

☐ Solar charging confirmed (midday check)

☐ Node visible in Meshtastic network list

QUICK-CARD 7 ◀ BACK

TROUBLESHOOTING & LOG FORMAT

Problem	Action
Not visible on mesh?	Check power, reboot, check antenna connection
Short battery life?	Check solar panel angle and cleanliness
LED off?	Check fuse, cable connections, battery voltage
Enclosure wet inside?	Check gasket seal; add desiccant pack

LOG ENTRY FORMAT

Date: _____	**Checked by: _____**
Battery V: _____	**Solar V: _____ Status: _____**
Notes: _____	

QUICK-CARD 8 — POWER CONSERVATION GUIDE

QUICK-CARD 8 ▶ FRONT

POWER CONSERVATION — EMERGENCY MODE

PRIORITY ORDER (use only what's available)

10. Essential life-safety communications **(always on)**
11. Net check-in radio **(scheduled times only)**
12. Mesh node **(duty cycle: 50%)**
13. Information gathering / monitoring
14. Non-essential devices **(OFF)**

RADIO POWER SAVINGS

- Transmit only when necessary
- Use **lowest power setting** with reliable contact
- Monitor only between scheduled transmit times
- Share charging among group members

QUICK-CARD 8 ◀ BACK

ESTIMATED BATTERY DRAW

Device	RX	TX / Active
Baofeng UV-5R	~150 mA	~1.6 A
Meshtastic node	~60–100 mA average	
Raspberry Pi 4	~600 mA – 1.2 A	

Device	RX	TX / Active
LED lantern	~50–200 mA	
Smartphone charging	~500 mA – 2 A	

RULE OF THUMB:

Amp-Hours = (Device mA

÷

1000)

×

Hours

Example: 100 mA

×

24 h =

2.4 Ah

per day per node

QUICK-CARD 9 — OFFLINE DATA ACCESS

QUICK-CARD 9 ▶ FRONT

ACCESSING OFFLINE RESOURCES

ON YOUR USB KIT (insert USB drive):

Folder	Contents
MANUALS/	Field manual, radio guides
MAPS/	Offline map files (OsmAnd / Maps.me)
SOFTWARE/	Meshtastic flasher, Briar APK, Kiwix
MEDIA/	Training videos, quick-card PDFs
DATA/	Contact directory, frequency lists
TEMPLATES/	Message forms, log sheets

OFFLINE WIKIPEDIA

15. Install Kiwix app (in SOFTWARE folder)
16. Open Kiwix → Add library → Browse USB → open .zim file
17. Full Wikipedia available offline

QUICK-CARD 9 ◀ BACK

OFFLINE MAPS & FIRMWARE FLASH

OFFLINE MAPS — QUICK START

App	Steps
OsmAnd	Files → Import → Select map file from USB
Maps.me	Menu → Download maps → My files → USB

MESHTASTIC FIRMWARE FLASH

18. Connect device via USB-C
19. Open Meshtastic Flasher (in SOFTWARE folder)
20. Select your **device model**
21. Click **Flash** — wait 2 minutes
22. Pair via Bluetooth with Meshtastic app

QUICK-CARD 10 — MUTUAL AID PRINCIPLES

QUICK-CARD 10 ▶ FRONT

MUTUAL AID — CORE PRINCIPLES

#	Principle
1	**SOLIDARITY, NOT CHARITY** — We help each other as equals
2	**NO GATEKEEPING** — Access is not means-tested
3	**MOST VULNERABLE FIRST** — Prioritize those with greatest need
4	**TRANSPARENT DISTRIBUTION** — Account for all shared resources
5	**NO HIERARCHY IN CRISIS** — Anyone can ask for and give help
6	**DOCUMENT EVERYTHING** — Written records protect everyone
7	**SELF-CARE IS COMMUNITY CARE** — Helpers must also rest

QUICK-CARD 10 ◀ BACK

RESOURCE REQUEST PROCESS

23. State the need **clearly and specifically**
24. Verify with the requesting person / household
25. Check available resources
26. Fulfill from **nearest source**
27. Document: what was given, to whom, when
28. Replenish stock when possible

NEVER deny based on:

identity, ability to pay, relationship to the group, or perceived "worthiness." Access is unconditional.

QUICK-CARD 11 — AFTER-ACTION REVIEW (AAR)

QUICK-CARD 11 ▶ FRONT

AFTER-ACTION REVIEW — TEMPLATE

Conduct within **24 hours** of any exercise or incident.

Question	Notes
WHAT WAS PLANNED? *(Intended outcome)*	
WHAT HAPPENED? *(Factual description)*	
WHAT WENT WELL?	1. _____ 2. _____ 3. _____

QUICK-CARD 11 ◀ BACK

AAR — IMPROVEMENT & ACTION ITEMS

WHAT NEEDS IMPROVEMENT?

1. _____ 2. _____ 3. _____

ACTION ITEMS

Item	Owner	Due Date

Item	Owner	Due Date

WHO WAS PRESENT: _____
DATE: _____ FACILITATOR: _____
NEXT REVIEW DATE: _____

QUICK-CARD 12 — GO-BOX ESSENTIALS

QUICK-CARD 12 ▶ FRONT

GO-BOX — DEPLOY CHECKLIST

☐ VHF/UHF radio + programmed frequencies

☐ HF radio (if equipped) + antenna + tuner

☐ Meshtastic device + charged battery

☐ Antenna(s) + coax cables + adapters

☐ Power: battery bank + solar panel + cables

☐ USB kit (32 GB) + laptop / tablet

☐ Logbook + pens + message forms

☐ Quick-cards (this set, laminated)

☐ License copy + FCC authorization

☐ Headset / earphones for noise environments

QUICK-CARD 12 ◀ BACK

SETUP & TEARDOWN ORDER

SETUP ORDER

29. Select **elevated, open** location
30. Set up **antenna first** (before radio)
31. Connect power source; verify voltage
32. Power on radio — check frequency
33. Monitor before transmitting
34. Check in on local net or calling frequency
35. Establish and maintain **communications log**

TEARDOWN

☐ Log final entry (time, reason for teardown)

☐ Disconnect antenna **before** moving radio

☐ Secure all cables before transport

☐ Verify USB kit is packed

These quick-cards are part of the Offline Network Resilience System.

Print. Laminate. Distribute freely.

Version 1.0 — 2026 — **CC BY-SA 4.0** — 12 Cards Total

No internet required. No registration required. Share this document with anyone who needs it.